

Redwan Khalifa

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Summary of Qualifications

Computer Engineering student with embedded systems experience in C/C++, Assembly, Python, Git/GitHub, and hardware-software integration across firmware, PCB, and real-time validation workflows. Developed and debugged embedded systems using SPI, I2C, Linux, oscilloscopes, and logic analyzers while supporting CI/CD-enabled testing and technical documentation. Strong fit for Agile embedded controls development through microprocessor, digital systems, signals, and software architecture experience.

Education

Bachelor of Engineering (BEng) Computer Engineering
Toronto Metropolitan University (CEAB/ABET Accredited)

Expected - 2026
Toronto, ON

- Minor in Computer Science, Mathematics, Biology and certification in ASL (American Sign Language).
- Relevant Coursework: Electronic Circuits, Digital Systems Engineering, Embedded Systems, Microprocessor Systems, Computer Architecture.

Experience

Software Engineer

01/2026 - Present

Riipen

Toronto, ON

- Translating a Figma UI/UX prototype into a mobile application using swift, implemented full stack features while ensuring usability and performance.
- Designed AI-powered automation features to streamline data validation, and workflow processes, reducing manual effort and improving feature delivery.

Avionics Team Member

06/2025 - Present

MetRocketry

Toronto, ON

- Contributed to a full-stack telemetry GUI using Electron, React, Node.js, Python Flask, and a CI/CD pipeline for automated builds/tests.
- Designed an SQLite relational database for real-time flight data visualization, and added validation layers that reduced parsing errors by 60%.
- Collaborated on a 4-board circular modular PCB system in KiCad/Altium, integrating SMD sensors and MCU bus protocols for telemetry and power delivery.
- Migrated firmware from Teensy 4.1 to 4.0, reducing bay footprint by 30%, and upgraded systems from I2C to SPI for faster, more reliable communication.
- Modernized firmware architecture with C/C++, Python, Roff and Python-based flashing/testing tools, cutting code redundancy by 30%.

Full Stack Web Developer

02/2025 - 05/2025

Riipen

Toronto, ON

- Led front-end feature development in HTML/CSS/JavaScript, improving load speeds by 40% and contributing to a 54% increase in user engagement.
- Built and integrated a backend-powered contact form, increasing successful submissions by 35% and reducing client response delays by 25%.
- Deployed web features using a CI/CD pipeline, cutting deployment time by 60% and reducing post-release bugs by 40%.
- Integrated RESTful APIs to synchronize real-time data between client and server, improving reliability and cross-device performance by 30%.

Private STEM Tutor

04/2023 - 12/2025

Independent Contractor

Toronto, ON

- Delivered personalized 1-on-1 and small-group STEM tutoring to 15+ middle-school, high-school, and university students, improving grades by 15-25%.
- Developed individualized study plans and progress reports, helping students strengthen problem-solving confidence and reduce assignment errors by 30%.

Industrial Machine Operator

05/2021 - 05/2023

Tradition Fine Foods

Toronto, ON

- Operated industrial-scale mixing bowls, tray washers, drum lifters, dispensers, and ingredient dumpers within a high-throughput production environment.
- Performed troubleshooting, equipment adjustments, and line support while handling 50 lb materials and maintaining efficiency along conveyor system.

Projects

TMU - Smart Campus Navigation System

[GitHub](#)

09/2025 - Present

- Engineered the backend architecture using Node.js, Express, and MongoDB to deliver real-time indoor navigation with <100 ms average query time.
- Implemented Dijkstra's algorithm enabling multi-floor shortest-path routing across 30+ rooms using 10 interconnected Bluetooth beacons.
- Integrated Firebase Authentication, improving platform security, adding support for TMU emails and ensuring protected access for 100% of user sessions.
- Led backend specification design, creating API schemas and data-flow models that supported error-free integration across three subteams and scaled to handle 10,000+ routing requests daily.

Digital Signal Processing Filter Design and Audio Equalizer

[GitHub](#)

08/2025

- Implemented DTFT, DFT, and FFT algorithms in Python to analyze discrete signals, improving spectral evaluation accuracy by 30%.
- Designed and tested an FIR high-pass filter; validated frequency response through spectral plots with <5% error across target bands.
- Built a 3-band audio equalizer (bass/mid/treble), demonstrating applied DSP concepts and achieving smooth ± 12 dB gain control with stable output.

Skills

Hardware: I2C, SPI, Verilog, VHDL, Quartus, Xilinx ISE, KiCad, Altium, Arduino, Raspberry Pi, HCS12, AutoCAD.

Programming: Python, C/C++, C#, Java, Assembly, Roff, MATLAB, Bash.

Lab: Oscilloscopes, Logic Analyzers, Spectrum Analyzers, Multimeters, PCB prototyping, soldering Iron.

OS & Tools: Windows, Linux, MacOS, Android, iOS, Git/GitHub, Jupyter Notebook.

Databases: Microsoft SQL Server, MySQL, PostgreSQL, SQLite, NoSQL, MongoDB, MongoDB Atlas.

Canadian Citizen

Open to Relocation

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